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Accreditation Council for
Graduate Medical Education

Exploring the Alignment of Clinical Cases Conducted during Training and into Early Practice Using ACGME Case Log and ABNS POST Data Systems

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MISSION

Encourage the Study,
Improve the Practice,
Elevate the Standards,
and Advance the
Science of Neurological
Surgery and Thereby to
Serve the Cause of
Public Health.

MISSION

Improve health care and
population health by
assessing and enhancing
the quality of resident
and fellow physicians'
education through
advancements in
accreditation and education.

Disclosures & disclaimers

None of the investigators has financial conflicts of interest.

Opinions expressed are those of the investigators and may not reflect the policies of the ABNS or ACGME

From residency to certification

YEARS 1 TO 7

NEUROSURGERY RESIDENCY

Complete ACGME-
accredited
neurosurgery
residency program

- Neuroanatomy exam
- Primary Exam: pass for credit prior to chief year

YEARS 7 TO 10

FELLOWSHIP

Enfolded or
post-residency
Optional 1-2
years
specialized
training

PRACTICE

ABNS POST[®]
Case Log

YEAR 10-14

ORAL EXAM

Case log approved
Application:
verification of
training and letters
of reference.
Exam: standard
questions and your
own cases

BOARD CERTIFIED

Professionalism
General/emergency
neurological surgery
adaptive learning
tool
Improving health
and health care

CONTINUING CERTIFICATION

Resident

ABNS Candidate: Board Eligible/Tracking Toward Certification

Diplomate

ACGME Case Log data



- Trainee NPI, PGY
- Date & role played
- Procedure (CPT codes, technique)
- Mandatory minima varies by procedure type
- Self entered, unaudited

Case logs help ensure that programs provide each trainee with sufficient volume of varied procedural experiences.

They do not ensure competence.



POST Case Log data

- Surgeon NPI
- Demographics
- Practice type & descriptions
- Fellowship type
- Procedure (CPT codes, technique, & text)
- Self entered, unaudited
- Mandatory log at least 100 cases.

ABNS are intended to help understand early-career neurosurgeons thinking, learning, adaptability to outcomes.

ABNS POST® Case Logs



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Within 3 years of completing training, 100 case log and application due

Major surgical cases involving 100 consecutive unique patients

Neurosurgeon is the primary surgeon (case is billed under the neurosurgeon's NPI) and performs pre and postop care

Reoperations or staged operations are bundled by medical record number

At least 3-month follow-up

All cases performed within 24 months of submission

Minor case log: separate spreadsheet

ACGME Case Log Data

Defined Case Category	Required Minimum Number	
Cranial	Senior + Lead Cases*	Lead Cases*
Cranial: Tumor General	60	30
Cranial: Tumor Sellar/Parasellar	20	10
Cranial: Trauma/Other	60	30
Cranial: Vascular Open	10	--
Cranial: Vascular Endovascular	10	--
Total Cranial Vascular	60	30
Cranial: CSF Diversion/ETV/Other	20	10
Cranial/Extracranial: Pain	10	5
Cranial/Extracranial: Functional Disorder	10	5
Cranial/Extracranial: Epilepsy	10	5
Total Cranial	300	150
Spinal	Senior + Lead Cases	Lead Cases
Spinal: Anterior Cervical	30	15
Spinal: Posterior Cervical	30	15
Spinal: Thoracic/Lumbar/Sacral/Instrumentation/Fusion	30	15
Spinal: Lumbar Laminectomy/Laminotomy	30	15
Spinal: Stimulation/Lesion/Pump/Other	10	5
Total Spinal	300	150
Peripheral Nerve	10	5
Radiosurgery	10	5
Peripheral Device Management	20	10
Critical Care	Senior + Lead Cases	Lead Cases
Airway Management	--	10
Angiography	--	20
Arterial Line Placement	--	10
CVP Line Placement	--	10
EVD/Transdural Monitor Placement	--	30
Lumbar/Other Puncture/Drain Placement	--	10
Percutaneous Tap of CSF Reservoir	--	10
Total Critical Care	--	100
Pediatric***	Senior + Lead Cases	Lead Cases
Pediatric: Cranial Tumor	5	-
Pediatric: Cranial Trauma/Other	10	5
Pediatric: CSF Diversion/ETV/Other	10	5
Pediatric: Spinal	5	-
Total Pediatric	40	20
TOTAL ALL DEFINED CASE CATEGORIES	800	400
Intradural Microdissection**	--	80

Cranial
(ACGME/POST)

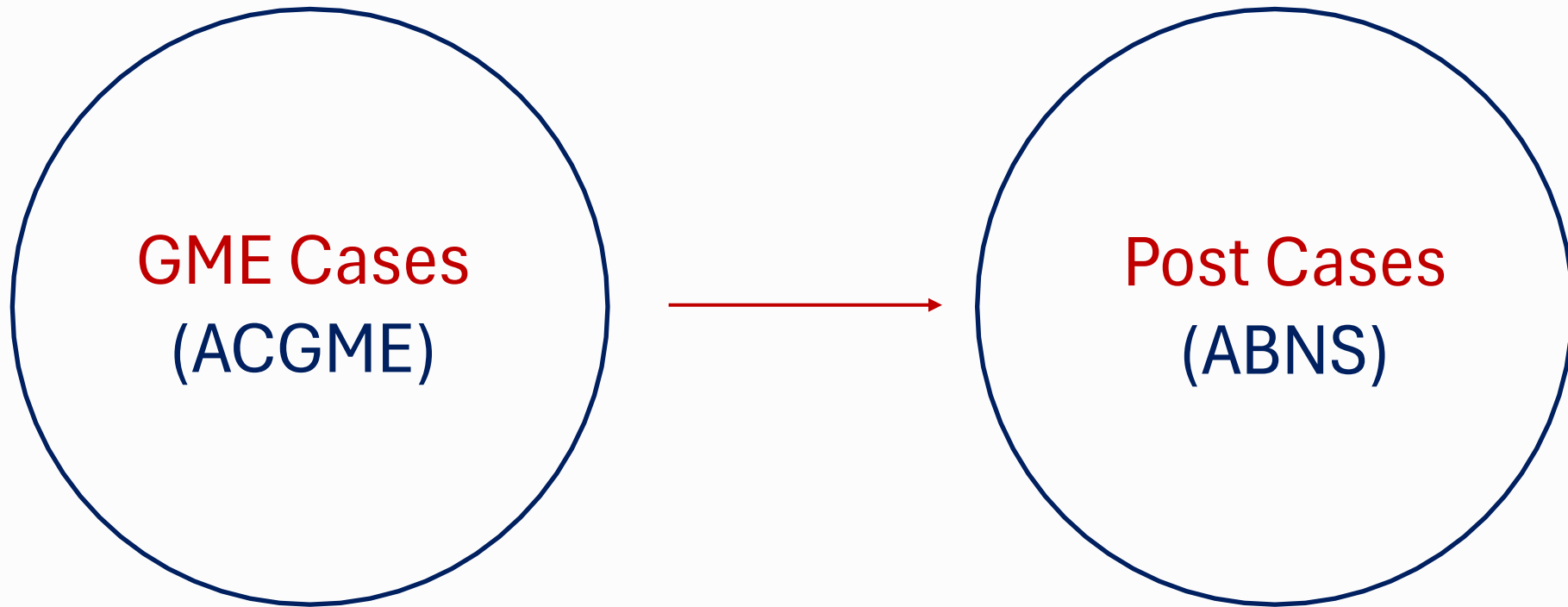
Spinal
(ACGME / POST)

Pediatric
(ACGME / POST)

Peripheral
Nerve (POST)

Critical Care
(ACGME)

Research Questions – Analysis



1. What are the prevalence of cases performed by neurosurgeons during residency (GME/ACGME) and in early practice (POST/ABNS)?
2. What is the association between AGME cases and POST cases?
3. Are associations moderated by practice setting?

POST Data

POST data: $n = 1,659$ unique NPIs (207,273 records)

ACGME Case Log Data

- Merge % with ABNS POST data: **74.4%** ($n = 1,235$)

	Graduation Year	Merged for Analysis		Post Data		Population
		n	%	n	%	n
Study Sample $n = 949$	2014-2015	68	89.5%	76	45.2%	168
	2015-2016	103	98.1%	105	56.5%	186
	2016-2017	119	98.3%	121	65.1%	186
	2017-2018	170	100.0%	170	91.4%	186
	2018-2019	151	100.0%	151	95.0%	159
	2019-2020	185	99.5%	186	95.9%	194
	2020-2021	166	96.0%	173	86.9%	199
	2021-2022	158	98.8%	160	74.8%	214
	2022-2023	92	98.9%	93	41.3%	225
Total		1,212	98.1%	1,235	71.9%	1,717

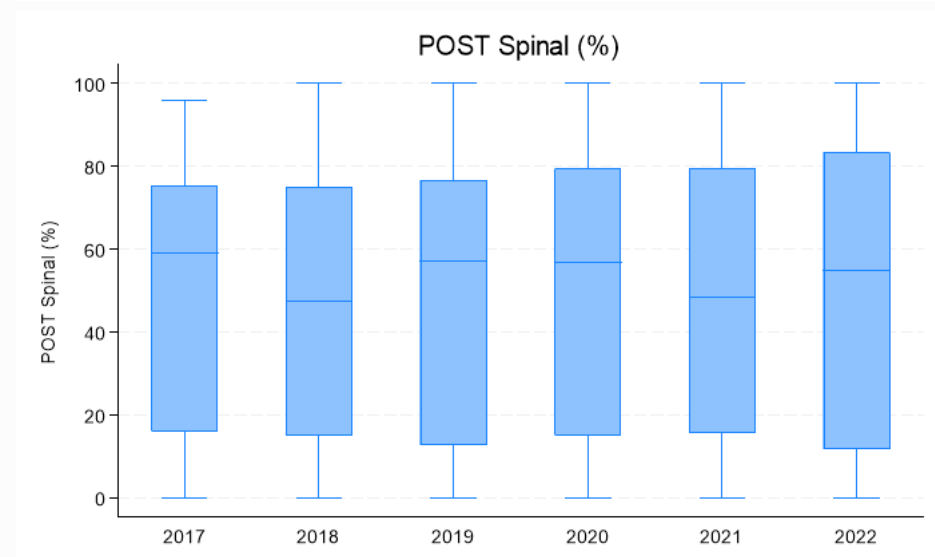
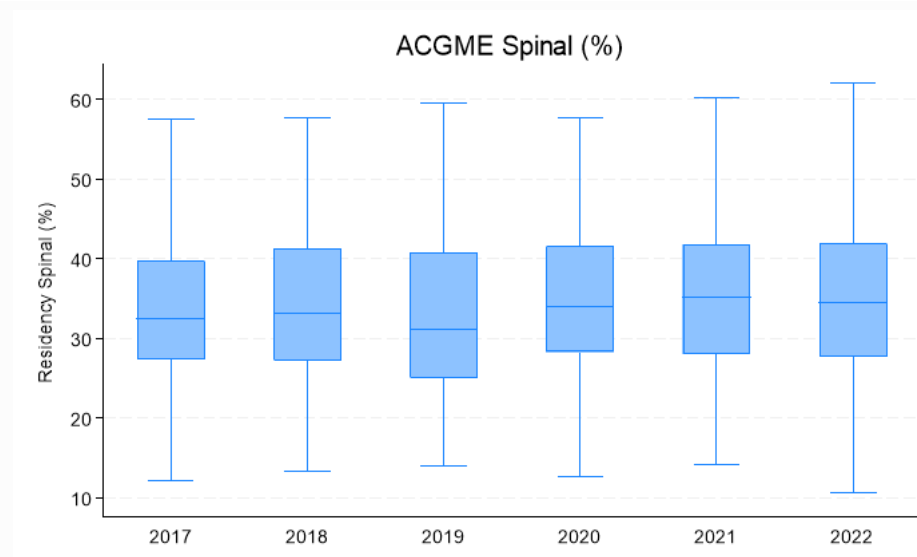
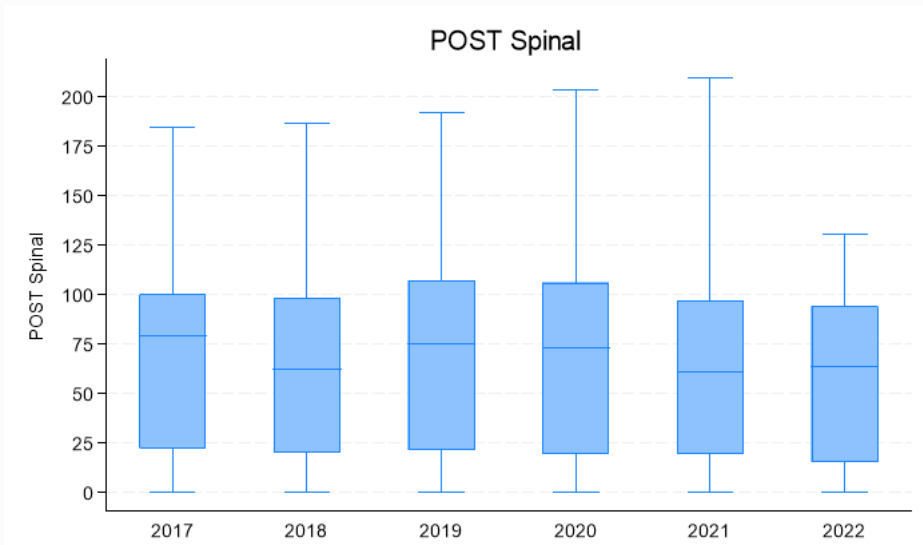
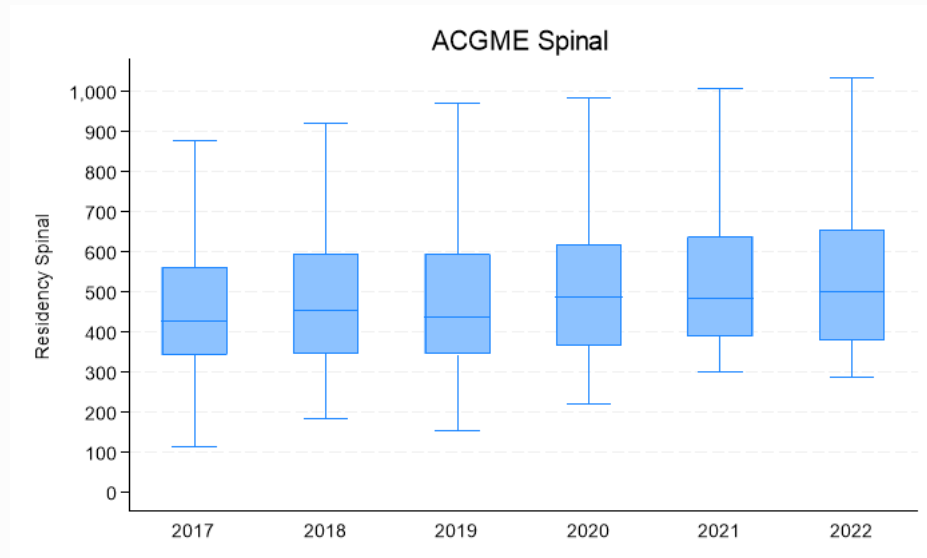
Proportion Cases (%): Descriptive Statistics (Graduates 2017 – 2022: 6 years; $n = 949$)

		Mean	Median	P-value
Spinal	ACGME	34	33	<0.001
	ABNS	48	54	
Cranial	ACGME	45	44	<0.001
	ABNS	41	32	
Pediatric	ACGME	7	6	0.91
	ABNS	8	1	

Note: Proportion (%) relative to total cases reported

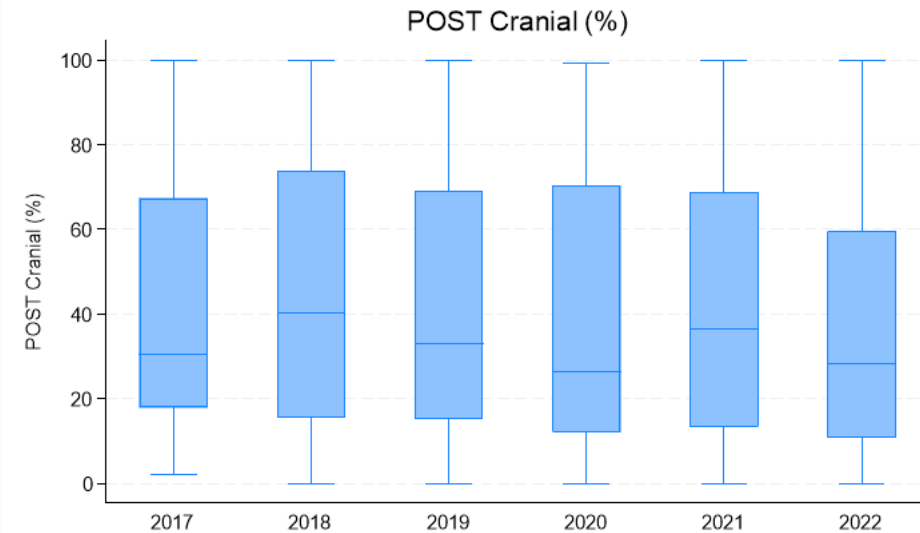
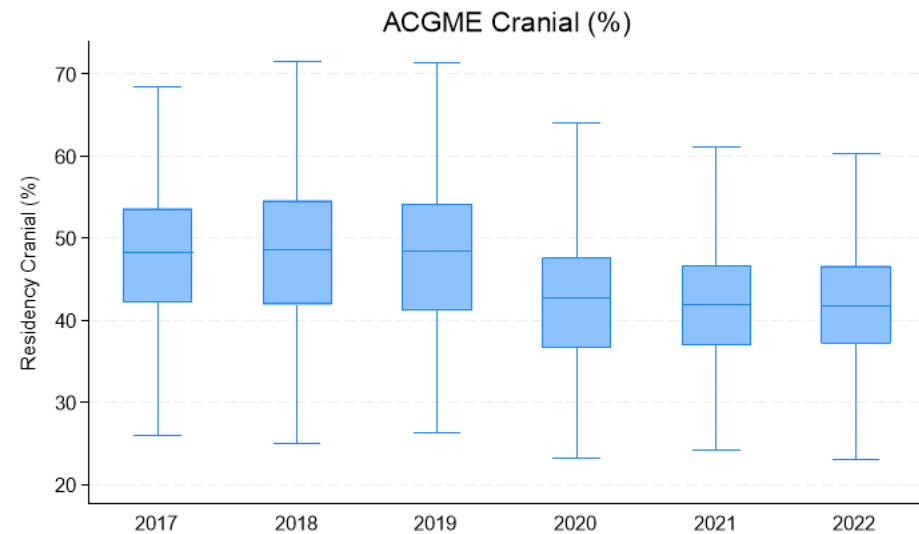
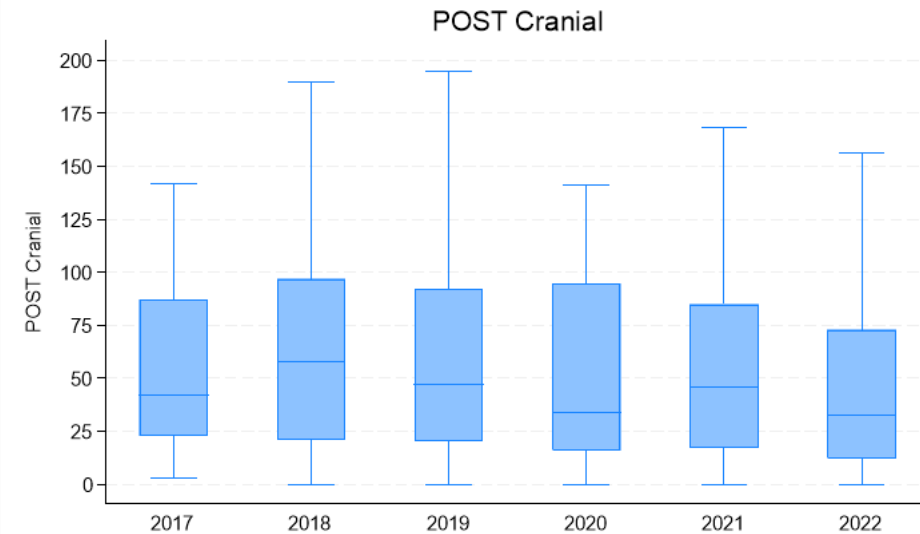
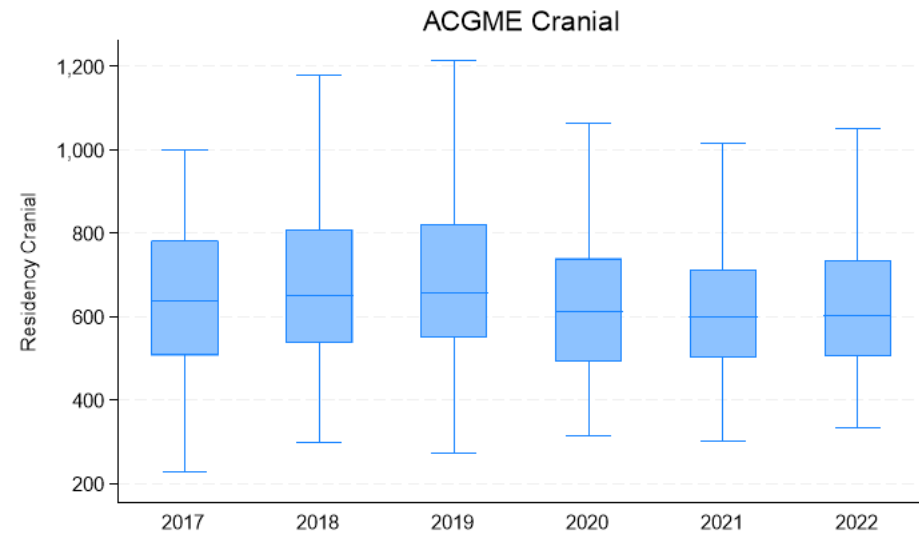
Spinal Cases

(Graduates 2017 – 2022: 6 Years)



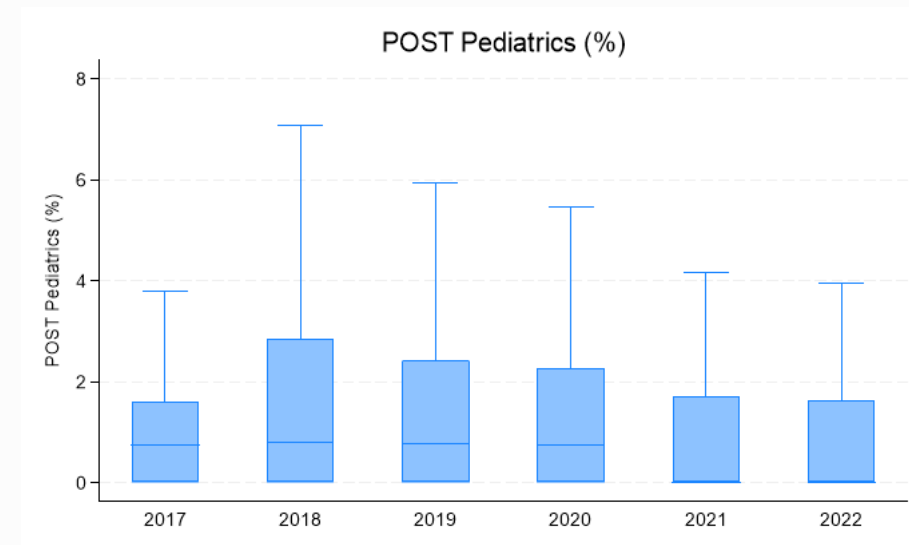
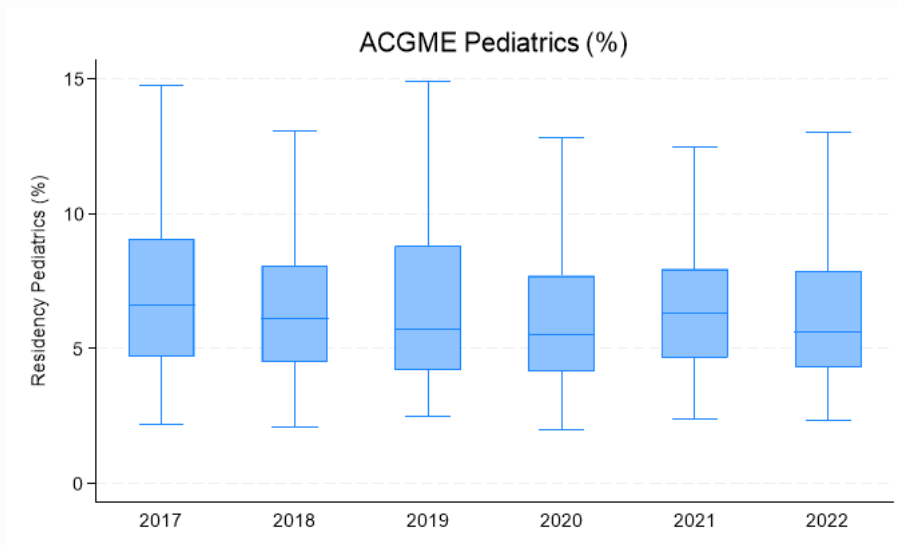
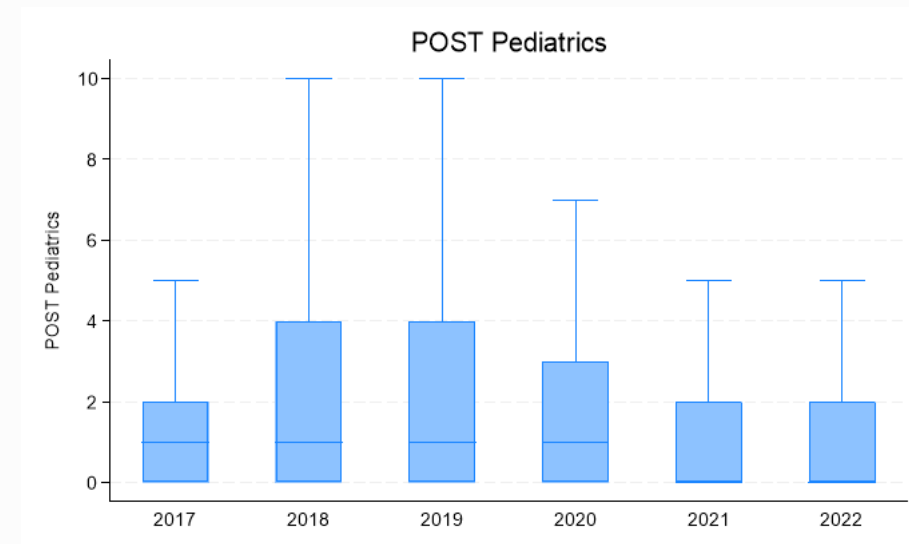
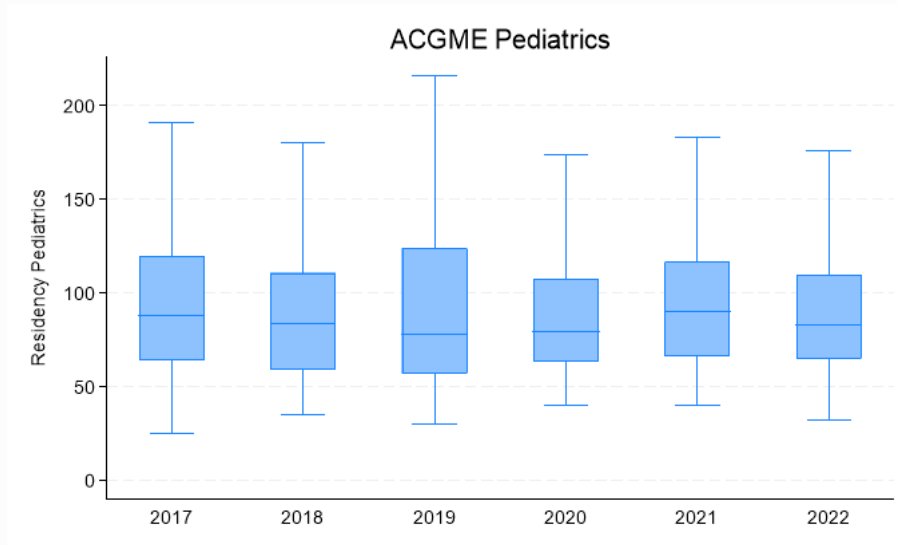
Cranial Cases

(Graduates 2017 – 2022: 6 years)



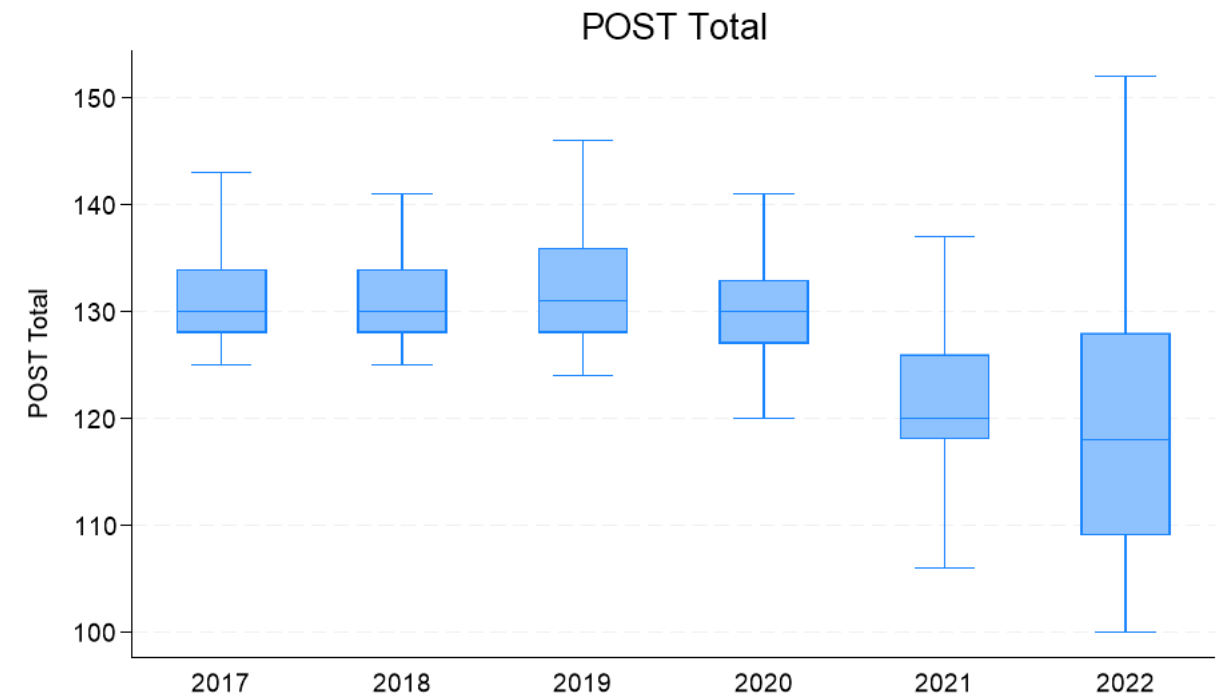
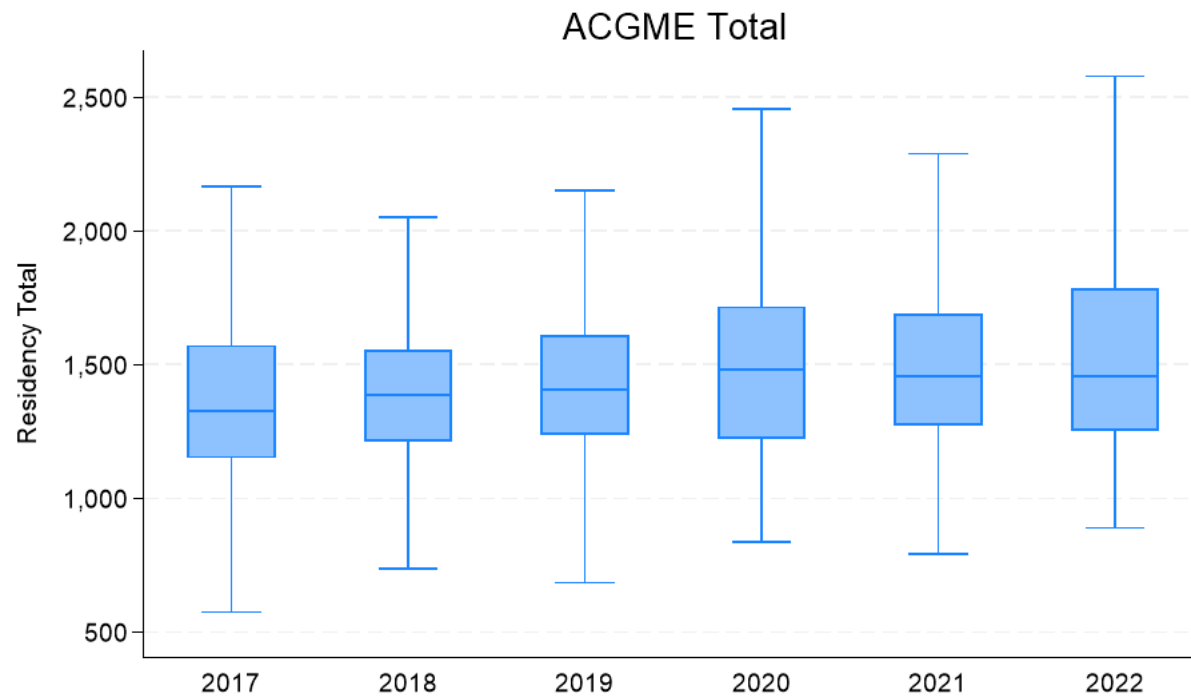
Pediatric Cases

(Graduates 2017 – 2022: 6 years)

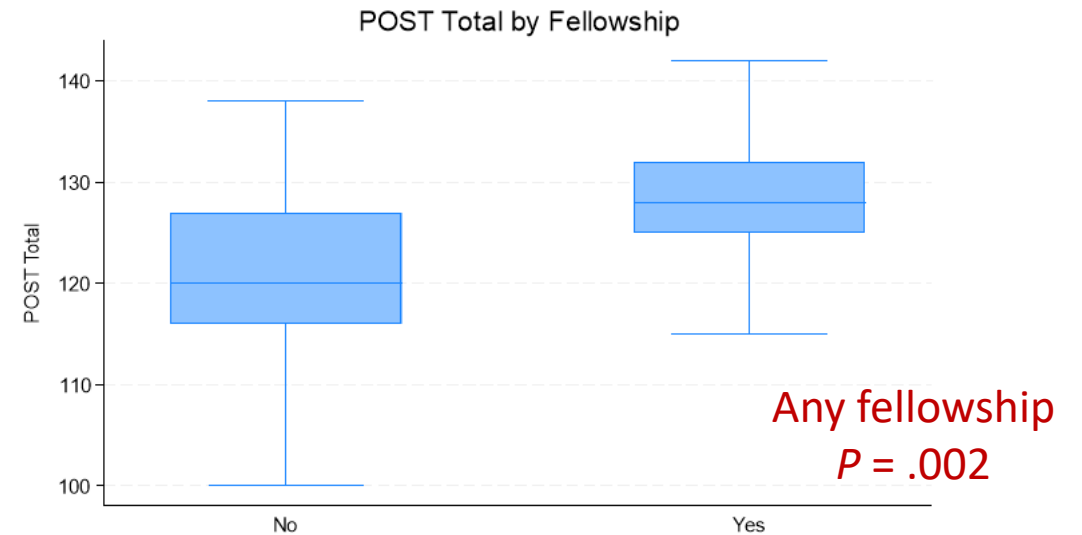
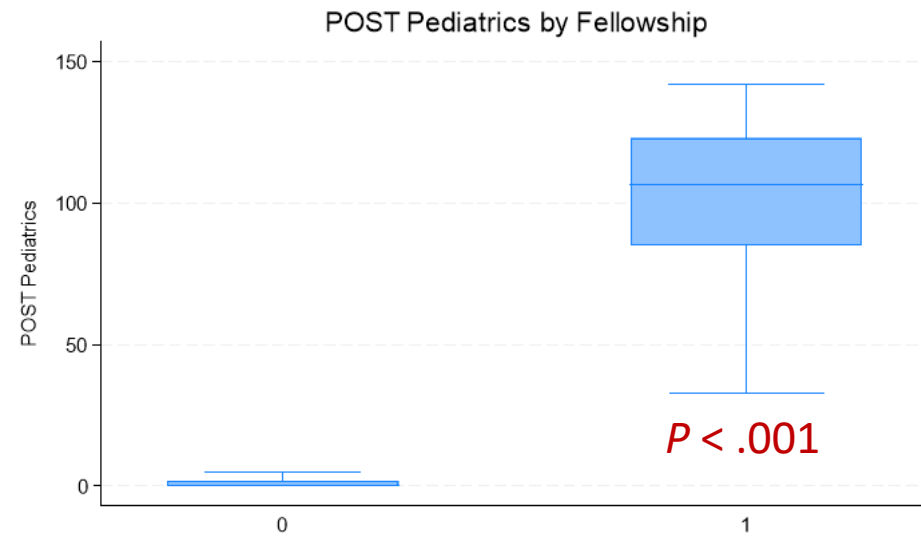
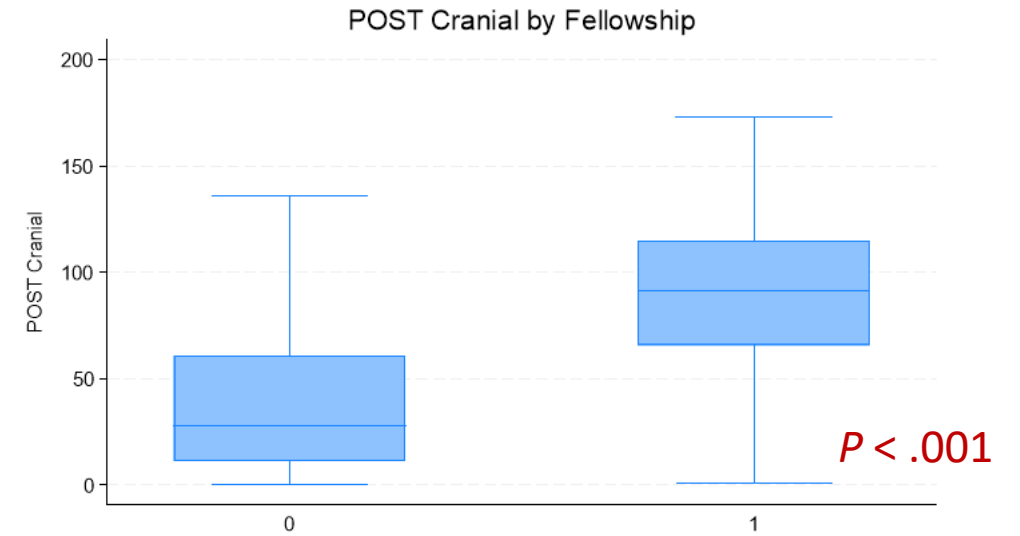
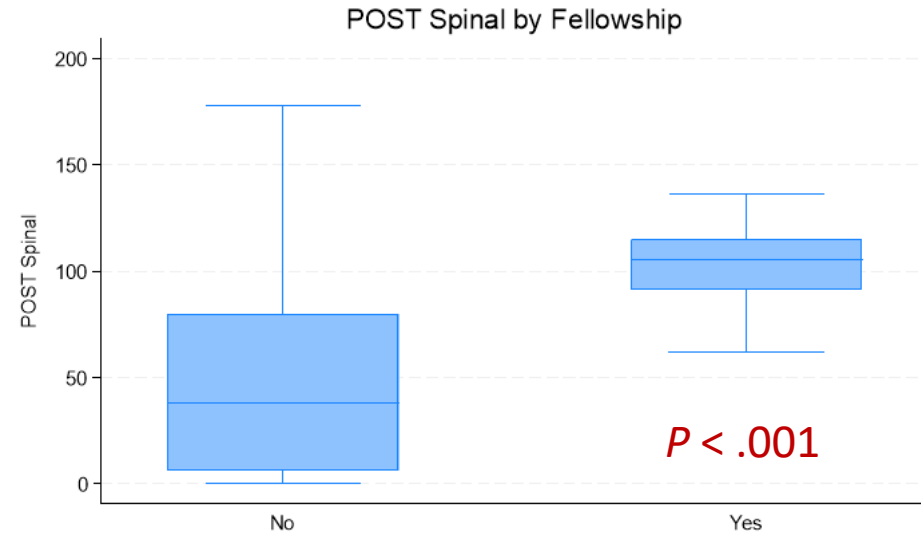


Total Cases

(Graduates 2017 – 2022: 6 years)



POST cases by Fellowship Status



Associations: ACGME ↔ POST

(Graduates 2017 – 2022: 6 years)

Proportion of cases in location		Spinal	Cranial	Pediatric	p-value
Description	General	0.30	0.36	0.09	ns
	General w specialty	0.53	0.51	0.22	<.001
	Specialty	0.68	0.58	0.39	<.001
Type	Academic	0.61	0.55	0.27	<.001
	Hospital	0.48	0.49	0.37	<.001
	Private	0.48	0.43	0.27	<.001
Overall		0.56	0.51	0.29	<.001

Conclusions

- This is the first correlation of residency surgical case volume to post-training case volume
- The types of cases performed in training closely reflect cases performed in early practice
- Practice setting is associated with case type concentration
- Findings suggest the potential to use data like these to calibrate training and initial-certification requirements

Discussion